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Autoencoders in Multi-Omics Data Integration

**Introduction to Bioinformatics
Section: A**

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1 Introduction

Modern biology produces huge amounts of data from different sources, such as:

- Genomics (DNA)
- Transcriptomics (RNA)
- Proteomics (proteins)
- Metabolomics (metabolites)

They are called **multi-omics data** in together. Analyzing this type of data is difficult because:

- The datasets are extremely large
- Different omics types behave differently
- Some data is often missing
- Relationships between variables are very complex

Traditional statistical methods struggle with these challenges. Because of this, researchers now use deep learning methods, especially **autoencoders**, to analyze multi-omics data.

Autoencoders are powerful because they can:

- Reduce data size
- Learn hidden patterns
- Combine different omics datasets
- Handle noisy or incomplete data

This report explains how autoencoders and their advanced versions are used in multi-omics research.

2 What is an Autoencoder?

An **autoencoder** is a type of neural network that learns how to compress data and then reconstruct it.

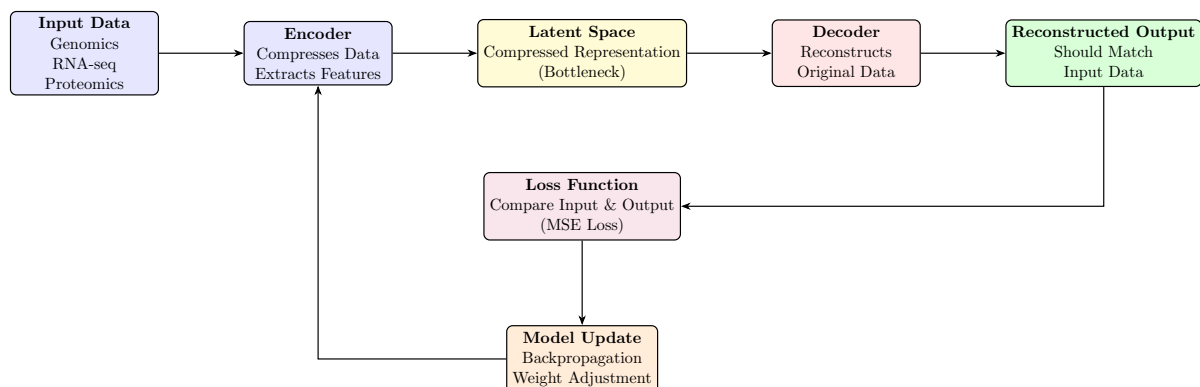


Figure 1: Basic Workflow of an Autoencoder

It has two main parts:

- Encoder: The encoder takes large complex data and converts it into a smaller simplified form called a **latent representation**.
- Decoder: The decoder tries to rebuild the original data from this compressed version.

The goal is to keep the most important information and to remove unnecessary Noises.

2.1 How Autoencoder Works

Working Process of an Autoencoder

- The encoder takes high-dimensional omics data as input.
- It learns important biological patterns from the data.
- The data passes through several neural network layers.
- Each layer reduces the size of the data step by step.
- Unimportant and noisy information is removed.
- Important features are compressed into a smaller representation.
- This compressed representation is called the **latent space** or **bottleneck**.
- The latent space stores the most useful biological information.
- The compressed data is then sent to the decoder for reconstruction.
- In multi-omics, the encoder helps combine genomics, transcriptomics, proteomics, and epigenomics into one unified representation.

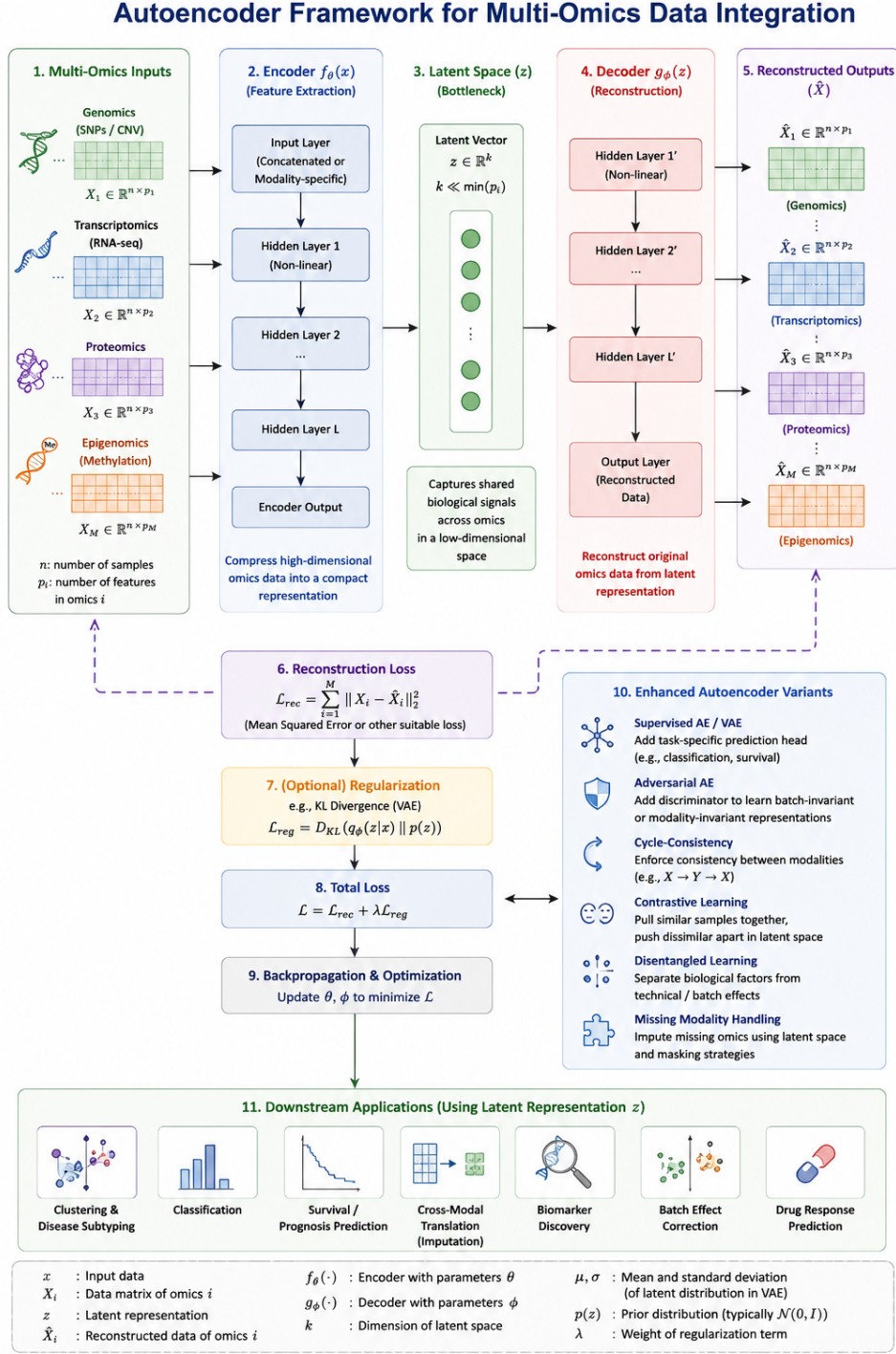


Figure 2: Autoencoder Architecture

2.2 Why Autoencoders are Useful in Multi-Omics

Multi-omics datasets contain thousands or millions of biological features. Autoencoders help in ways:

- They find key patterns in biology
- They combine different types of data from various sources

They support tasks such as

- Predicting Diseases
- Identifying different types of cancer
- Grouping similar things together
- Filling in missing data

Unlike older methods like PCA autoencoders can understand complex relationships, which makes them more useful for biological data. Autoencoders are really good, at handling data because they can learn non-linear relationships .

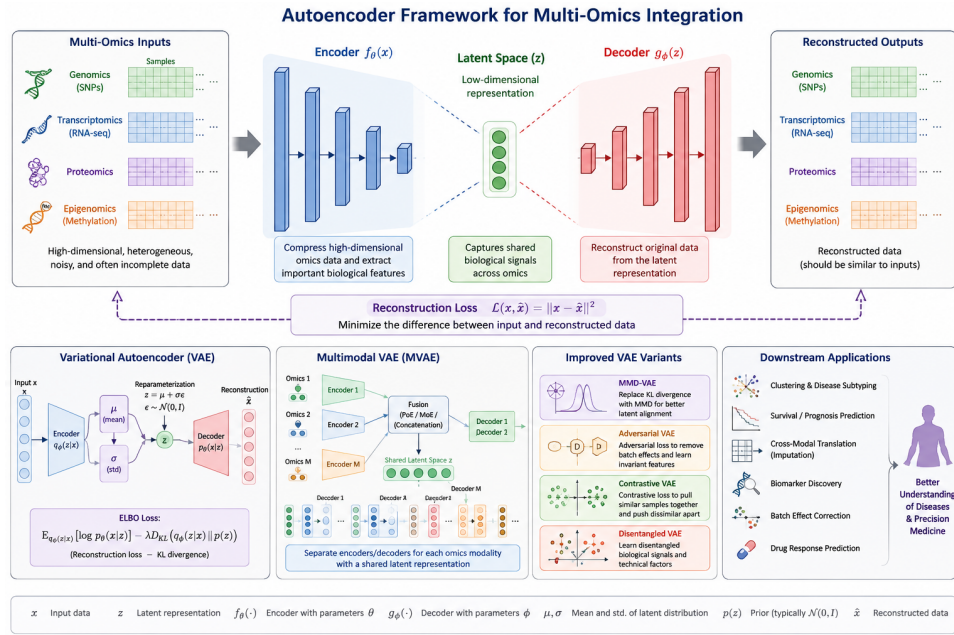


Figure 3: Autoencoder Architecture

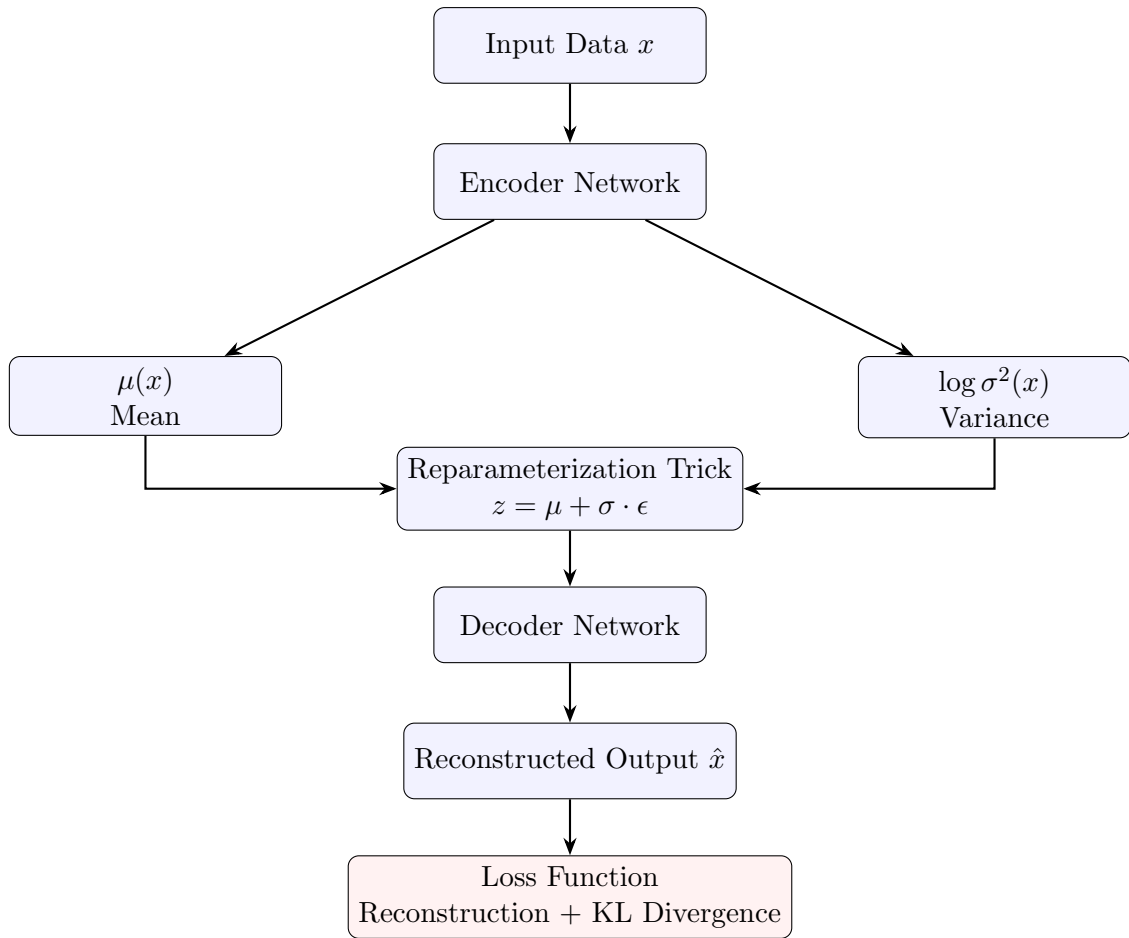
3 Variational Autoencoders (VAEs)

A Variational Autoencoder or VAE for short is like a supercharged version of an autoencoder.

It does not just learn a compressed value. Instead it learns a range of possible values like a probability distribution.

This is really useful because:

- The model gets a sense of how sure or unsure it's about things.
- It can create brand synthetic data that is similar to what it was trained on.
- The VAE helps to make the latent space, which's like a compressed version of the data, smoother and more meaningful.



3.1 How VAEs Work

In a VAE:

- The encoder makes:
 - Mean (μ)
 - Standard deviation (σ)

These tell us about a probability distribution.

The model picks a sample, from this distribution to make a hidden representation.

The decoder uses this sampled hidden vector to rebuild the data.

Variational Autoencoder (VAE) – Flowchart

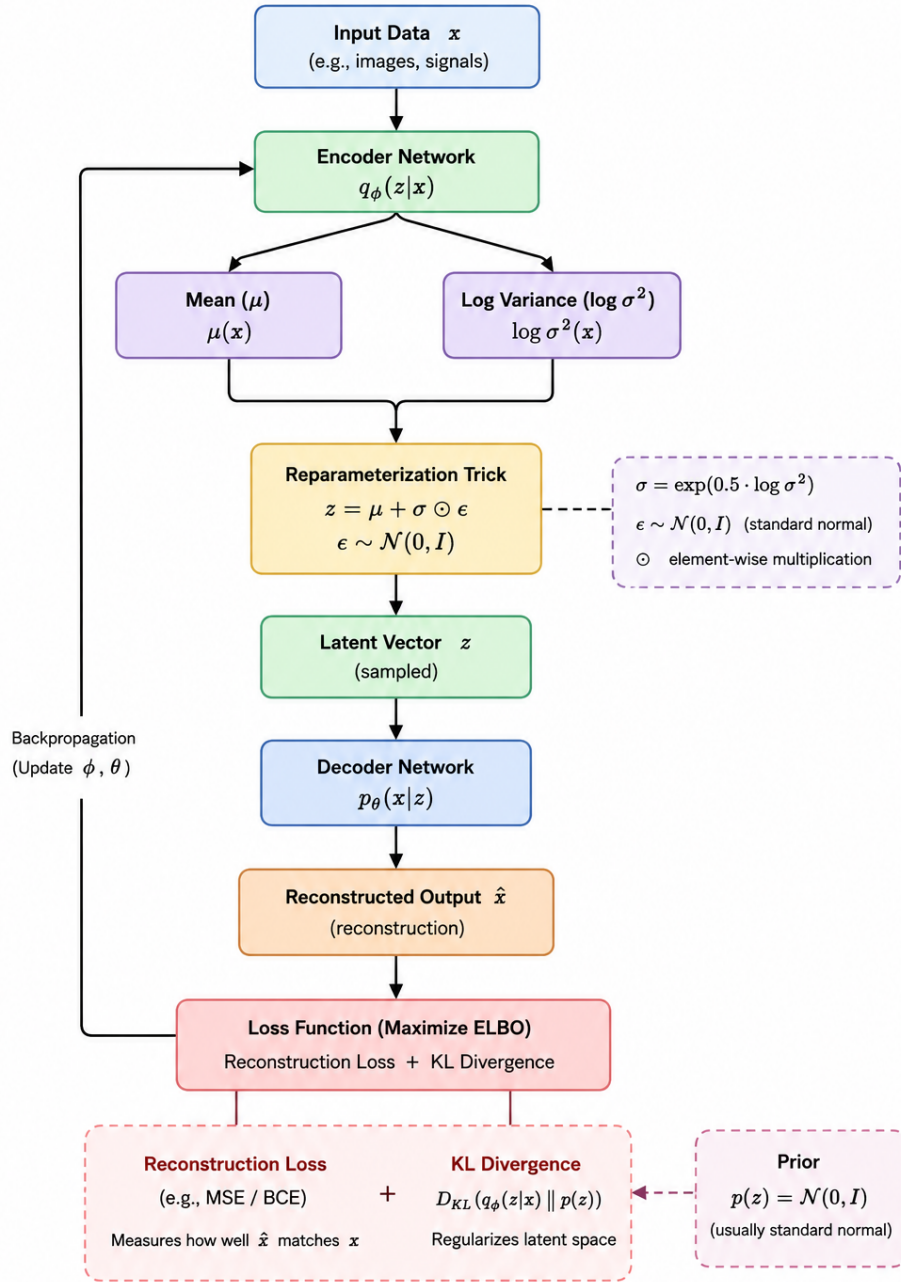


Figure 4: VAE Architecture

3.2 Why VAEs are Important

Variational Autoencoders or VAEs are really useful for multi-omics. They can do a lot of things.

- Handle noisy biological data: VAEs can handle biological data.
- Generate synthetic samples: This is important because biological datasets are often incomplete or inconsistent.

- Learn meaningful latent spaces: This enables us to find common biological signatures across omics types.
- Support uncertainty estimation: Important in medical and biological applications

4 Important Improvements in VAE-Based Models

Different enhancements to the method have been proposed to improve the performance of Variational Autoencoders (VAEs) in multi-omics data integration. These advancements are primarily aimed at boosting the quality of latent graphical representations, handling multimodal data, improving comprehension, and addressing missing data problems.

4.1 Regularization and Latent Space Enhancement

Posterior collapse is an important obstacle in VAEs where the model ignores the latent space.. Alternative regularization strategies have been recommended to this end. .

4.1.1 MMD Regularization

Maximum Mean Discrepancy (MMD) is used (or added) to KL divergence to improve latent structure learning.

Benefits:

- Improved latent representations
- More stable training process
- Better preservation of biological signals

Representative Models:

- MMD-VAE
- Multigrade
- MultiVI
- scDisInFact

4.2 Supervised and Semi-Supervised VAEs

The standard VAE is unattended, but biological datasets are frequently labeled with metadata such as disease type or clinical outcome. Combining supervision improves prediction performance.

Applications:

- Cancer classification
- Prediction of drug response
- Survival analysis

Benefits:

- Enhanced classification accuracy
- More effective interpretability for biological systems
- Enhanced disease prediction performance

Representative Models:

- OmiVAE
- XOmiVAE
- OmiEmbed

4.3 Multimodal Integration Strategies

Multi-omics data are obtained from several biological sources. Multimodal VAEs desire to fuse these heterogeneous datasets into a common latent modal representation.

4.3.1 Fusion Strategies

- Concatenation-based fusion
- Product of Experts (PoE)
- Mixture of Experts (MoE)

Benefits:

- Captures biological signals shared
- Enables cross-omics integration

Representative Models:

- scMVAE
- scMM
- Multigrade
- MultiVI
- MIDAS

4.4 Cross-Modal and Coupled Representation Learning

Instead of projecting all modalities into a common latent space, some models keep independent representations and discover the relationships between them.

Benefits:

- Preserves modality-specific information
- Learns cross-omics relationships

Representative Models:

- DCCA
- JAMIE

4.5 Adversarial Learning Approaches

Discriminator-based adversarial training aligns variations across standardized modalities or eliminates a batch effects. Each encoder learns to fool the discriminator and is better aligned with it.

Targets of Discrimination:

- Data modality
- Batch/source effects

Benefits:

- Better cross-modal alignment
- Batch effect reduction

Representative Models:

- SCIM
- GLUE
- scCross
- omicsGAN

4.6 Cycle-Consistency Learning

The cycle-consistency provides sure that data converted between different modes and back preserves the original information.

Benefits:

- Improved cross-modal translation
- Better reconstruction consistency
- More reliable embeddings

Representative Models:

- scMVP
- sciCAN
- Portal

4.7 Contrastive Representation Learning

Contrastive learning is an illustration learning method that acquires knowledge by pulling together similar samples and pushing apart dissimilar samples.

Example principle:

- Same tissue samples → closer in latent space
- Different tissue samples → more distant

Benefits:

- More biologically meaningful embeddings
- Enhanced clustering and downstream prediction

Representative Models:

- MOSA

4.8 Disentangled Representation Learning

In biological data, there is often biological information as well as technical noise. These factors are separated by Disentangled methods in the latent space.

Benefits:

- Cleaner biological interpretation • Improved batch correction

Despite their capabilities, these models still suffer from issues. Representative Models: **Capable of predicting missing omics modalities in the system**

Despite their capabilities, these models still suffer from issues. Representative Models:

- scDisInFact
- MIDAS

5 Handling Missing Data in Multi-Omics Integration

In real-world datasets, not all samples contain all omics modalities. This setting is known as mosaic integration.

5.1 Common Strategies

- Masking missing modalities during training
- Zero-padding incomplete inputs
- Using prior-based imputation techniques

5.2 Benefits

- Enables learning from incomplete datasets
- Improves robustness in real-world clinical data
- Allows prediction of missing omics modalities

Representative Models:

- Multigrade
- scVAEIT
- MIDAS

6 Popular Autoencoder Models

Model	Main Idea	Main Use
OmiVAE	Supervised variational autoencoder with label guidance	Cancer classification
MMD-VAE	Uses Maximum Mean Discrepancy instead of KL divergence for better latent structure	Cancer subtype prediction
OmiEmbed	Multi-task VAE for joint representation learning	Survival prediction
scMVAE	Multimodal VAE with shared latent space	Clustering and imputation
Multigrade	Product-of-Experts based multimodal integration with MMD regularization	Mosaic multi-omics integration
MultiVI	Joint embedding across multiple omics modalities	Batch effect correction
GLUE	Graph-based generative model for regulatory network learning	Gene regulatory inference
MOSA	Contrastive learning-based representation model	Biomarker discovery
MIDAS	Disentangled representation learning framework	Batch correction and imputation

Table 1: Popular Autoencoder-Based Models in Multi-Omics Integration

7 Challenges

Even though these models are powerful, they still have problems.

7.1 High Computational Cost

Deep learning models need:

- GPUs
- Large memory
- Long training times

7.2 Limited Data

High-quality multi-omics data sets are costly and not readily available. Malfunctions due to small data sets can lead to:

It is often hard to explain the biological basis of deep learning models. Researchers still struggle to understand: The meaning of each of the latent dimension why the model make certain prediction

7.3 Interpretability Problems

Deep learning models are often difficult to explain biologically. Researchers still struggle to understand:

- What each latent dimension means
- Why the model makes certain predictions

7.4 Hyperparameter Tuning

The following factors are critical to the performance: • Latent space size • Loss weights • Network design It's hard to find the optimum settings.

8 Future Directions

Researchers are now exploring:

8.1 Foundation Models

Large pretrained biological AI models such as: • scGPT • scFoundation

8.2 Integration Beyond Omics

Future systems can be a mixture of: • Multi-omics data • Medical images • Electronic health records • Wearable sensor data This could greatly improve precision medicine

8.3 Better Missing Data Handling

Future models will better support: • Incomplete datasets • Large patient populations • Real-world clinical data

8.4 Better Interpretability

Researchers are looking for models that are: • More transparent • Easier to explain • More clinically trustworthy

9 Conclusion

Autoencoders and Variational Autoencoders (VAEs) are important deep learning models that are used for The integration of multi-omics data for efficient representation learning from high dimensional data. and HeterogeneousBiologicalData. They are common to use for the following: Compress and visualize large-scale omics data • Add one or several modalities of biological data to the model Assist in the prediction and stratifying of diseases • Identify and resolve missing or incomplete data • Model non-linear biological processes. To enhance performance for real-world biological applications, several additions have been made to the main program.A number of additions are made to the main program to improve performance for real-world biological applications. Various methods have been developed, such as MMD-based regularization, adversarial training, contrastive learning, disentanglement methods, and multimodal fusion strategies. These advances have led to better biologically-relevant robustness in these models. The incorporation of VAEs into today's computational biology is a must-have. Their The further development of the research is expected to further advance precision medicine, biomarker discovery, and tailored health care.